

From Pair Kernels to Higher-Order Information

The Strict Loop-Product Flag Lift,
Hidden-Synergy No-Go and Adaptive Source Obstruction

FIN ST2172–ST2261

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

Six adaptive research rounds — 90 programmes
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This report tests the highest-value frontier left by Release 11.07: whether the strict FIN pair kernel, its states, adaptive law, or fractal refinement can source a mathematically unavoidable ternary/2-form object. It identifies one previously underclassified positive result—the strict loop-product flag lift—and immediately subjects it to symmetry, probabilistic, adaptive, refinement, Hodge-spectral, kernel-split and axiom-removal falsification.

Abstract

Let $A = sI - W$ be the strict twelve-state Dirichlet generator, with positive off-diagonal conductances W_{ij} . Strict pair data generate a canonical reducible three-cycle statistic

$$\tau_{ijk} = W_{ij}W_{jk}W_{ki}.$$

All 220 triangle values are positive, lie in twelve D_{12} -orbits and define a target-free weighted flag-complex candidate. This corrects the coarser earlier statement that the pair kernel generated no intrinsic three-cycle observable.

The correction does not close the physical bridge. The statistic τ is fully reducible to pair data and reflection-even. A centered Gaussian theory generated by a quadratic FIN action has vanishing connected third cumulants. The functional-calculus algebra $C^*(A)$ is commutative and associative. State-dependent chiral currents exist, but the $k = \pm 1$ branches have equal energy, opposite current and are exchanged by reflection. Invariant mixtures, D_{12} -twirls and Gibbs states have zero current.

The adaptive law $\dot{K} = \Pi[C - V'(K)]$ does not repair the source problem without additional data. Spectral-algebra flows cannot generate a noncommuting or odd structure. Self-generated dephased covariance commutes with A , while its origin comes from preparation. The strict W is indefinite, so it cannot equal a positive covariance divided by positive decay in the elementary linear Hebb fixed-point equation.

An exact information-theoretic obstruction is established using

$$p_\varepsilon(x_1, x_2, x_3) = \frac{1}{8}(1 + \varepsilon x_1 x_2 x_3), \quad |\varepsilon| \leq 1.$$

Every member has the same complete one- and two-point records, while $\mathbb{E}[x_1 x_2 x_3] = \varepsilon$. Marginalization erases the distinction. No deterministic reconstruction from pair/coarse data can therefore recover general irreducible ternary information.

The positive loop field yields an executable Hodge family

$$h_f(p) = \frac{\tau_f^p}{\langle \tau^p \rangle}, \quad p > 0.$$

Every member is positive, target-free and D_{12} -natural. The choices $p = 1$ and $p = 2$ give different one-form spectra with the same cohomology. Current strict data therefore do not select unique gauge dynamics.

The smallest conditional uniqueness theorem needs four extra axioms: triangle-edge locality, support-zero behaviour, separate linearity in each edge coupling, and one universal face rule. They imply $F(a, b, c) = \kappa abc$. Removing any one admits explicit counterfamilies; explicit permutation symmetry is redundant. None of the four is yet derived as a physical law.

The updated gate passes three strict mathematical rows and zero physical-source rows. FIN gains a rigorous higher-order graph lift, not a sourced 2-form physical theory. The next decisive theorem is to derive or refute separate linearity and face universality from the strict composition/refinement law.

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1 Scope and kernel discipline

The legacy and strict objects remain separated:

$$K_{\text{legacy,ont}}(d) = \alpha_{\text{geo}} \frac{\cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d},$$

$$K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^n}.$$

No legacy physical role is transferred. The strict Dirichlet representation used here is

$$A = sI - W, \quad \langle f, Af \rangle = \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0.$$

Release 11.07 conflated two meanings that must be separated:

- (i) a three-cycle statistic derived from pair data;
- (ii) irreducible third-order information not determined by pair data.

The first exists. The second is not generated by the current pair core.

2 Round I: ST2172–ST2186 — third-order classification

The frozen strict matrix is invariant under all 24 elements of D_{12} ; the finite conjugation residual is zero.

Theorem 1 (fixed-input odd-scalar no-go). *If $rx = x$ for a reflection r , and $F(rx) = -F(x)$, then $F(x) = 0$.*

Proof. $F(x) = F(rx) = -F(x)$. □

This applies to a global pseudoscalar derived equivariantly from the fixed pair kernel. It does not exclude oriented cochain bases or state-dependent odd receivers.

For a centered Gaussian field, Wick's theorem gives

$$\langle \phi_i \phi_j \phi_k \rangle_c = 0.$$

Linear unitary, wave and heat evolution preserve Gaussianity. Thus the dual dynamics

$$U_t = e^{-itA}, \quad P_t = e^{-tA}$$

do not create connected third cumulants from Gaussian data.

All Borel functions of one self-adjoint A commute, and matrix multiplication is associative:

$$[f(A), g(A)] = 0, \quad (f(A)g(A))h(A) - f(A)(g(A)h(A)) = 0.$$

The polynomial replay residuals were 2.73×10^{-13} and 4.23×10^{-14} ; the exact facts follow from functional calculus.

Proposition 1 (strict loop-product field).

$$\tau_{ijk} = W_{ij}W_{jk}W_{ki}$$

is positive on all 220 strict triangles:

$$9.2543082 \times 10^{-5} \leq \tau_{ijk} \leq 4.2419834 \times 10^{-2}.$$

Its support values form exactly twelve D_{12} -orbits.

This is a local closed-walk statistic. It is intrinsic as a graph observable but is reducible to (W_{ij}, W_{jk}, W_{ki}) , reflection-even, and not a connected third cumulant.

3 Round II: ST2187–ST2201 — state chirality

Let

$$\psi_k(n) = 12^{-1/2} e^{2\pi i k n / 12}, \quad \rho_k = |\psi_k\rangle\langle\psi_k|.$$

For $k = \pm 1$,

$$\langle\psi_{\pm 1}, A\psi_{\pm 1}\rangle = 0.75412115420708.$$

The state-kernel current

$$\Lambda(\rho, W) = \sum_i 2 \operatorname{Im}(\rho_{i,i+1} W_{i+1,i})$$

gives

$$\Lambda(\rho_{+1}, W) = -0.4699856726450202, \quad \Lambda(\rho_{-1}, W) = +0.4699856726450202.$$

Reflection exchanges the two states with residual 8.42×10^{-16} .

Theorem 2 (invariant-state persistence). *If Φ is D_{12} -covariant and ρ is D_{12} -invariant, then $\Phi(\rho)$ is invariant; every reflection-odd expectation remains zero.*

The branch mixture has zero current. The full twirl gives -3.26×10^{-18} , and the Gibbs state gives zero. The pitchfork $\dot{q} = (1 - q^2)q$ has stable branches $q = \pm 1$ but does not select one. Symmetric noise gives equal branch probabilities; a bias imports the missing odd premise.

The P2718 chiral bispectrum and P3048 memory-phase triangle area remain real odd receivers. P3048 has five nonzero signed rows but zero accepted strict source rows. Chirality exists at state/receiver level, not as a selected strict source.

4 Round III: ST2202–ST2216 — adaptive source audit

The family

$$\dot{K} = \Pi[C - V'(K)]$$

becomes a law only after C, V, Π , initial state and time scale are supplied.

Theorem 3 (commuting adaptive manifold). *If $C(K), V'(K) \in C^*(K)$ and Π preserves this algebra, the flow remains in $C^*(K)$. It cannot generate a noncommuting associator or odd source.*

The dephased covariance of self-generated unitary motion is block diagonal in the spectral projections of A . For a localized preparation it has rank seven and

$$\|[A, C]\|_\infty = 3.36 \times 10^{-16}.$$

Translating the prepared delta changes it by Frobenius distance 0.40824829; the apparent origin is preparation data.

For

$$\dot{W} = \eta(C - \gamma W), \quad \gamma > 0,$$

the only fixed point is $W = C/\gamma$. Covariance is positive semidefinite, whereas the strict W has

$$\lambda_{\min}(W) = -0.6818747624, \quad \lambda_{\max}(W) = 1.6603072788.$$

The elementary covariance-Hebb law therefore cannot learn strict W as its fixed point. This is not a no-go for all nonlinear laws. Reverse-engineering V can make a chosen K stationary, but infinitely many potentials do so; representation is not derivation.

5 Round IV: ST2217–ST2231 — exact hidden synergy

Theorem 4 (pair-sufficiency obstruction). *For*

$$p_\varepsilon(x_1, x_2, x_3) = \frac{1}{8}(1 + \varepsilon x_1 x_2 x_3), \quad |\varepsilon| \leq 1,$$

one has

$$\mathbb{E}[x_i] = 0, \quad \mathbb{E}[x_i x_j] = 0 \ (i \neq j), \quad \mathbb{E}[x_1 x_2 x_3] = \varepsilon.$$

No deterministic function of the complete one- and two-point record can recover the general third moment.

Proof. The binary-cube monomials are orthogonal characters. All pair records are identical, while the required output varies continuously. $\square$

For $\varepsilon = \pm 0.75$, fine total-variation distance is 0.75. Marginalizing any variable makes the pair distributions identical and the distance zero. Marginalization is coassociative, so order cannot recover the erased synergy.

Conditionally, this formalizes the internal-layer intuition: a fine state can carry ternary synergy invisible to a pair-limited coarse observer. It does not derive a physical observer or actual FIN probability law.

6 Round V: ST2232–ST2246 — flag-Hodge candidate

All 66 strict conductances are positive; the minimum is 0.0110708173. The support graph is K_{12} , whose clique complex is the full 11-simplex with 220 triangles.

Set

$$h_f(1) = \frac{\tau_f}{\langle \tau \rangle}, \quad L_1(h) = d_0 d_0^* + d_1^* \text{diag}(h) d_1.$$

The $p = 1$ candidate has 33 distinct eigenvalues at eight decimal places and

$$0.3480858052 \leq \lambda \leq 60.2493296805.$$

Theorem 5 (positive-power Hodge counterfamily). *For every $p > 0$,*

$$h_f(p) = \frac{\tau_f^p}{\langle \tau^p \rangle}$$

is positive, target-free, D_{12} -natural, and has the same flag support. All members have $H^1 = 0$, but their positive spectra vary.

For $p = 1, 2$,

$$\|\lambda(1) - \lambda(2)\|_\infty = 24.78961896,$$

with relative ℓ^2 distance 0.3484. Mean-one normalization leaves the trace equal. Continuous positive multiplicative valuations have $g(x) = x^p$; refinement multiplicativity reduces the family but does not select p . Setting $p = \eta = 1.8$ would be an unproved transfer from kernel damping to face measure.

The complete support graph has diameter one and supplies no local wavevector, lattice spacing, Lorentz cone or Maxwell limit.

7 Round VI: ST2247–ST2261 — minimal theorem

Define the *strict loop-product flag lift*

$$W \mapsto (\text{Clique}(\text{supp } W), \tau_{ijk} = W_{ij} W_{jk} W_{ki}).$$

Theorem 6 (conditional uniqueness). *Suppose a face rule $F(a, b, c)$ has:*

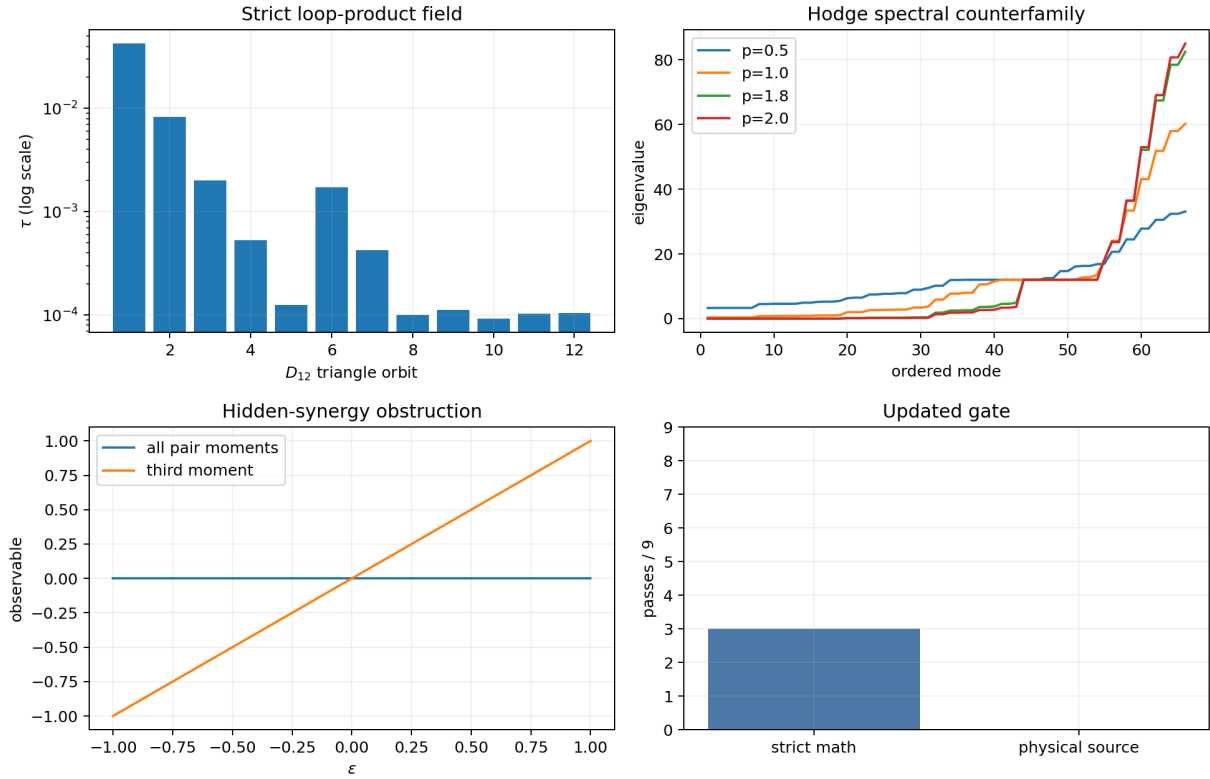


Figure 1: Loop-product orbits; the Hodge spectral counterfamily; hidden third-order synergy; and the updated mathematical-versus-physical gate.

- A1.** *triangle-edge locality;*
- A2.** $F = 0$ *if any support edge is absent;*
- A3.** *separate linearity in each edge coupling;*
- A4.** *one universal rule on every face, rather than one coefficient per D_{12} -orbit.*

Then $F(a, b, c) = \kappa abc$. Mean normalization fixes descriptive κ .

Axiom	Necessity	Result of removal
A1 locality	necessary	arbitrary global invariant multipliers enter
A2 support-zero	necessary	constant, linear and pair terms enter
A3 separate linearity	necessary	$(abc)^p$ and general functions enter
A4 universality	necessary	twelve orbit coefficients enter
explicit symmetry	redundant	κabc is already symmetric

These axioms give a small conditional theorem. A3 and A4 have not been derived from strict dynamics or refinement.

8 Updated gate and kernel-split falsification

Requirement	Strict mathematical	Physical source
reducible three-cycle observable	pass	fail
D_{12} law and nonzero witness	pass	fail
support-clique face complex	pass	fail
irreducible connected ternary source	fail	fail
nonpremise odd polarity	fail	fail
unique positive Hodge law	fail	fail
unique $12 \rightarrow 24 \rightarrow 48$ ternary lift	fail	fail
dimensional local continuum scaling	fail	fail
OA platform and independent record	fail	fail

3/9 strict mathematical rows,	0/9 physical-source rows.
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Strict positivity does not transfer to legacy. Raw legacy triangle products range from -12.3069 to 8.84166 , with 108 positive and 112 negative faces. A legacy positive Hodge model requires an additional absolute value, square, sign sector or completion choice. No bridge or legacy role transfer follows.

9 Result ledger

Statement	Status	Scope
strict positive three-cycle field	[Proven]	reducible statistic
Gaussian core generates third cumulant	[Refuted]	centered/linear
$C^*(A)$ supplies associator	[Refuted]	one generator
chiral state current	[Proven]	prepared receiver
operator selects one chiral branch	[Refuted]	symmetric core
linear covariance Hebb learns strict W	[Refuted]	positive decay
pair data reconstruct ternary synergy	[Impossible]	exact theorem
weighted flag-Hodge model	[Constructed]	dimensionless
one strict Hodge law is selected	[Refuted]	power family
four axioms select τ	[Conditional]	A1–A4
continuum gauge physics	[Not established]	scale/OA absent

10 Deepest surviving interpretation

The unavoidable higher-order shadow of the current strict FIN kernel is a reducible loop-product lift of a weighted graph into its flag complex. It is not an irreducible three-point information law. The operator supports conditional discrete Hodge/gauge models but does not select their valuation exponent, orientation sector, refined lift, physical units, continuum limit or operational realization. FIN remains a finite Dirichlet–Markov–spectral information geometry with a newly explicit higher-order graph lift, not a closed fundamental physical theory.

11 Next decisive programmes

The highest-value next theorem is to derive or refute A3 and A4 from the already strict composition/adaptive/refinement law, without importing the loop product or $p = 1$.

Recommended ST2262–ST2276:

ST2262 classify strict composition laws implying separate edge linearity;

ST2263 test whether strict symmetry can force face universality;

ST2264 classify continuous refinement valuations and exponent freedom;

ST2265 audit edge-subdivision consistency of τ ;

ST2266 compute $12 \rightarrow 24 \rightarrow 48$ loop-product renormalization;

ST2267 construct a minimal hidden-synergy instrument;

ST2268 test a sourced non-Gaussian adaptive third cumulant;

ST2269 classify third cumulants under reflection-equivariant noise;

ST2270 test strict selection of one chiral Fourier branch;

ST2271 prove identifiability or nonidentifiability of p ;

ST2272 test parameter-free sparsification without a threshold;

ST2273 build a sign-safe legacy loop lift and list its extra axiom;

ST2274 test one typed completion atom on triangle products;

ST2275 write a conditional CA+SA+OA record model;

ST2276 update the gate and stop if neither A3 nor A4 is derived.

12 Reproducibility and nonclaims

Six result sets from `FIN_ST2172_ST2186_Results.json` through `FIN_ST2247_ST2261_Results.json` contain ninety packet hashes. `test_fin_st2172_st2261.py` verifies all hashes, cycle statistics, state degeneracy, adaptive no-go, hidden synergy, Hodge counterfamily, gate and figure.

This report does not derive an irreducible ternary state law, selector, polarity, unique Hodge measure, physical scale, continuum gauge theory, apparatus, record, legacy completion, legacy role transfer, *QW*-2191, Standard Model, gravity, L_{total} , or Theory of Everything.

References

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